Zhongyuan Wang

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Founding ML Scientist specializing in AI, LLM, agent, and unsupervised graph models. 20 patents. 5 enterprise customers across the US, India, and the Gulf.

Work Experiences

2024 – Present **Founding ML Scientist**, *RaptorX.AI, Remote*

- Built an unsupervised fraud detection engine from zero combining proprietary GNN models, behavioral autoencoders, explainability, and LLM investigation reports. Differentiator: unifies unsupervised detection, graph fraud-ring discovery, and audit-ready explainability in a single system
- Engineered thousands of features across 4 scenarios, processing 10M+ transactions, 590K+ entities, 2.8M edges. Novel edge-probability model across 30 relationship types eliminates dependency on fraud labels
- Achieved 328$\times$ candidate reduction (590K $\rightarrow$ 1,883 high-confidence alerts), 99.9% fraud-type specificity
- Converted PoCs into 5 signed enterprise customers across the US, India, and the Gulf: Payactiv (US), NSDL and Lenskart (India), Omantel and Thawani Pay (Oman); US deployment covers 460K+ users across 6,700+ employers
- Technical deep-dives to global investors (Venture Guides, Claypond Capital) from seed stage onward

2021 – 2024 **Senior Engineer, AI & Data Platform for Autonomous Driving**, *Lotus Tech (Geely Group), Hangzhou, China*

- Led a fully in-house video anonymization pipeline from data-collection vehicles (80+ TB/day) to data center. YOLOv5-based model with 90%+ recall on faces and license plates across 10M+ frames, ensuring GDPR/PIPL compliance
- Reduced latency 30% via GPU-accelerated inference at fleet scale; saved ~$1M/year by replacing third-party vendors entirely

2020 – 2021 **Independent ML Study & Graph-ML Projects**

- Self-directed study of graph machine learning: implemented GNN models for node classification and link prediction on public Kaggle datasets

2020 **Software Development Engineer, Financial Anti-Fraud**, *DataVisor, Shanghai, China*

- **Production Fraud Detection for Top-Tier Commercial Bank:**
  - Built an unsupervised loan application fraud model combining GNN and traditional ML, using hundreds of features spanning behavioral, device, third-party credit, and raw transaction data
  - Directly contributed to securing a $500,000 enterprise contract through technical leadership and client demonstration

2019 **UCL-DiDi Master Research Program**, *Intern, DiDi AI Labs, Beijing, China*
Supervisor: Prof. Jieping Ye (VP of DiDi Chuxing, IEEE Fellow).
Project: Leveraging Graph Neural Networks for User Travel Intent Prediction.
Selected as one of only 2 projects (out of 30+ candidates) across the entire UCL cohort for the DiDi AI Labs research program.
This work laid the foundation for applying GNNs to financial fraud detection at RaptorX.

Patents & Publications

Patents (inventor): 20 applications spanning graph-based fraud detection and LLM/agent systems, with counsel: 6 filed (US), 4 published (India), 10 in draft. Published (India) App Nos.: 202541024623, 202541045354, 202441079562, 202541032797.

Publications (first author; all under review):

- When the Tool Decides: LLM Agents Defer Blindly to GNN Tools (TMLR; arXiv:2606.14476)
- LLM Features Can Hurt GNNs: Concatenation Interference on Homophilous Graphs (TMLR)
- How Much Do Deep GNNs Actually Help? Decomposing Depth, Features, and Structure (NeurIPS 2026 E&D)

Technical Skills

ML / Graph	PyTorch, PyTorch Geometric (PyG), GNNs, autoencoders, anomaly detection, link prediction, fraud-ring detection, LLM-based investigation
Data & MLOps	Spark / PySpark, Kafka, ClickHouse, Redis, Parquet, Docker, Kubernetes, ArgoCD, CI/CD, FastAPI / gRPC, MLflow
Cloud	AWS (S3 / EC2 / ECR), Grafana, Prometheus, Loki

Education

2018–2019 **University College London, London, UK.**
MSc in Data Science and Machine Learning.
Thesis: Graph Convolutional Networks for user behavior prediction (jointly with DiDi AI Labs).
Achieved top 3% in group NLP research project among 300 students.

2015–2018 **King's College London, London, UK.**
BSc in Computer Science (Artificial Intelligence).